

Linguistic Decolonization: Conditioned Political Bias of Edge LLMs Across South Africa's Languages

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Ask a large language model about South African politics in English and in isiZulu, and it may answer differently. Prompted in English, frontier models tend libertarian-left [12]; in languages they have seen little text in during training, the same model drifts toward more authoritarian framings, plausibly through transfer learning from English-centric data [1, 2, 7]. South Africa makes this measurable and consequential: twelve official languages, very unevenly represented in LLM performance, a fact about available data, not the standing of a language or its speakers [3, 4]; a model that misstates a community's politics distorts equitable access to information. We measure the gap. Varying the prompt language across the Afrobarometer South Africa survey languages, we measure how far a model's elicited political position diverges from its speakers', across a range of small edge models, up to about 6 GB of GPU memory. In parallel we benchmark each model's per-language capability, and the central question is where this divergence is largest and whether language, model size or language-capability explain it.

Research question

Across South Africa's official languages, where does a small edge LLM's political position diverge most from people's, and is that divergence explained by its language capability or size [2, 12]? Targeted contributions:

- a public per-language map of where edge LLMs' politics diverge from people's;
- a South African LLM-language-capability informed explanation of that divergence, and;
- evidence on whether that divergence is explained by model size.

Design

Per language, four steps:

1. **Run the survey on the models.** Administer the Afrobarometer South Africa items [8] (professionally translated, nationally representative human baseline; SASAS [9] cross-check) to small edge models of differing scale, up to about 6 GB of GPU memory, prompting each in the target language. Round 9 covers the nine most-spoken languages and SASAS adds Tshivenda, leaving only isiNdebele without a baseline.
2. **Measure language capability.** Score each model's per-language capability with AfroBench [11] (subsuming sets such as IrokoBench [3]), extended for the languages it under-serves, notably isiNdebele and Tshivenda, using resources such as MzansiText [4, 5].
3. **Compare model to people.** Per language, the divergence between the model's elicited political position and its speakers' real attitudes is the measure, following work on meaningful evaluation of political values in LLMs [6] and on LLMs as survey simulators [10]; below a capability threshold the gap is a finding, not a forced score.
4. **Correlate.** Test whether the divergence tracks

language capability (step 2) and model size within this edge range [2, 12], collapsing toward English-anchored defaults where capability is low.

Where models default to English-anchored political sentiments in a language, the question for our South African partners arises, whether this calls for linguistic decolonization. A remedy then points beyond evaluation and analytics toward fine-tuning and retraining LLMs on local languages - which would already lay out a path to more equitable access to information, and a more representative political compass for South Africa's multilingual society.

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